

## Electronic Supplementary Information

**Recall, not verification, is the bottleneck when frontier LLMs elucidate molecular structures from real spectra**

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This document contains the Supporting Information figures referenced in the main article. Data, predictions, and the code that regenerates every figure are released with the manuscript; see *Data and code availability* in the main text.



Fig. S1: Study design: open multimodal data (IRExp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free core pipeline.

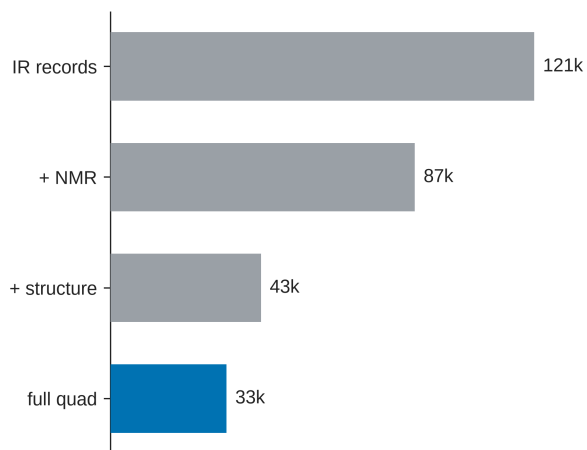


Fig. S2: IRExp composition: IR records, NMR-paired, structure-linked, and full IR+<sup>1</sup>H+<sup>13</sup>C+structure quadruples.

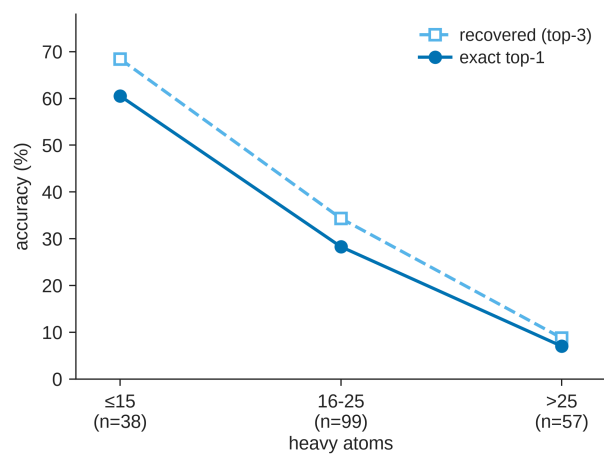


Fig. S3: Accuracy versus molecular size; the monotonic 60% $\rightarrow$ 7% top-1 gradient with heavy-atom count.

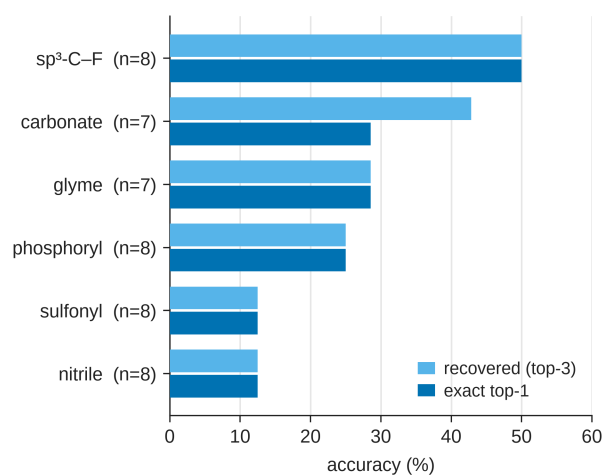


Fig. S4: IRSpectra-Bench-Electrolyte by battery-electrolyte class (n=46):  $\text{sp}^3\text{-C-F}$  easiest (50%), sulfonyl and nitrile hardest (12%).

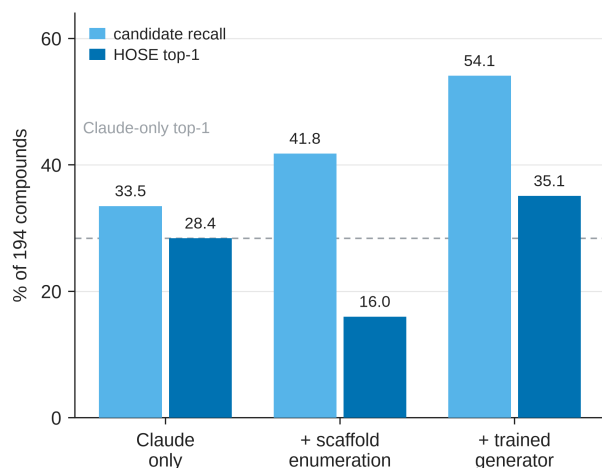


Fig. S5: Trained-generator probe (§5.6; a complement, not part of the training-free protocol). Candidate recall and deterministic-HOSE top-1 on the 194-compound benchmark for Claude / + scaffold enumeration / + trained generator: enumeration’s near-degenerate isomers collapse the verifier (28.4→16.0%) while the generator’s formula-correct candidates convert (28.4→35.1%).

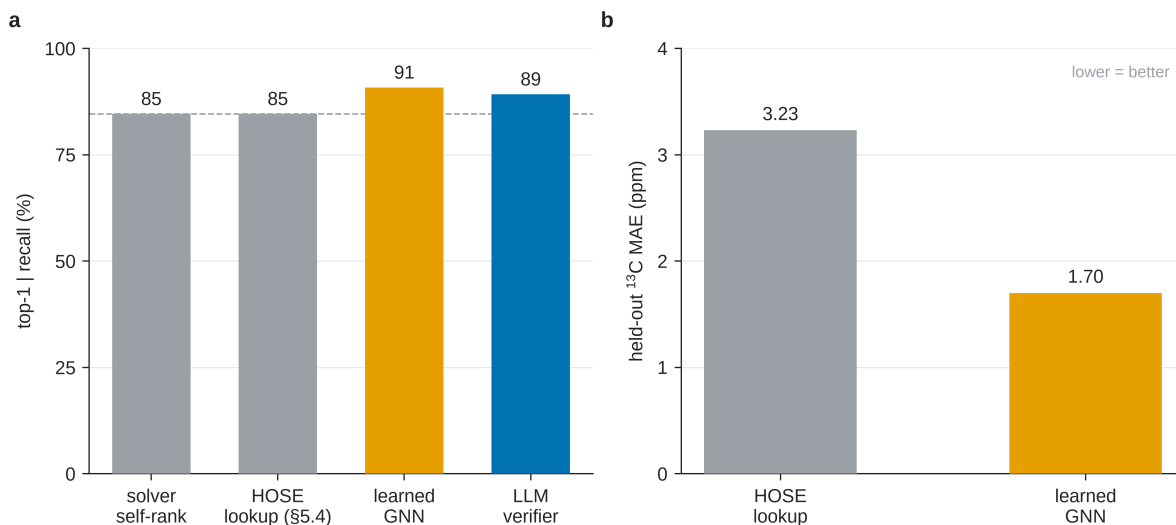


Fig. S6: Learned-verifier probe (§5.4; a complement, not part of the training-free protocol). (A) Conditional-on-recall top-1 over the whole benchmark (n=65) across four verifiers: a GNN trained on the same nmrshiftdb2 data as the HOSE lookup reaches the LLM verifier’s level (91% against 89%) where the lookup (85%) does not move off the solver’s own ranking. (B) Held-out <sup>13</sup>C MAE — the learned model is roughly 2× sharper (1.70 vs 3.23 ppm).